1.LIBRARY MANAGEMENT SYSTEM

1. AIM

To design and develop a basic Library Management System webpage using HTML. The webpage provides sections for librarian administration, book management, issuing and returning books, fine management, uploading study materials, user registration and login, book searching, borrowing history, profile details, and contact.

2. PROCEDURE

1. Open a text editor such as Notepad, Visual Studio Code, or any HTML editor.
2. Create a new file and save it as library_management.html.
3. Write the standard HTML document structure using <!DOCTYPE html>, <html>, <head>, and <body> tags.
4. Add a page heading and navigation links for Home, Admin, User, and Contact.
5. Create the Home section with a short description of the Library Management System.
6. Create the Admin section with forms for administrator login, adding books, issuing and returning books, recording fines, and uploading study materials.
7. Add an HTML table to display sample book details and Update/Delete buttons.
8. Create the User section with login and registration forms, book search, book details, borrowing history, fine information, study-material controls, and profile details.
9. Create the Contact section with name, email, and message fields.
10. Add a footer containing the copyright notice.
11. Save the file and open it in a web browser to view the webpage.
12. Test the navigation links and form controls. Note that this HTML-only page is a front-end demonstration; storing records, authenticating users, and performing real issue/return operations require JavaScript and/or a backend with a database.

3. PROGRAM CODE

The following is the HTML source code for the Library Management System webpage. It preserves the main sections and controls from the supplied code.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Library Management System</title>
</head>
<body>
    <header>
        <h1 align="center">Library Management System</h1>
        <hr>
        <nav align="center">
            <a href="#home">Home</a> |
            <a href="#admin">Admin</a> |
            <a href="#user">User</a> |
            <a href="#contact">Contact</a>
        </nav>
        <hr>
```

```
</header>

<section id="home">
    <h2>Welcome</h2>
    <p>The Library Management System helps students and librarians
    manage books, issue and return books, check fines, and download
    study materials efficiently.</p>
</section>
<hr>

<section id="admin">
    <h2>Librarian Admin</h2>
    <h3>Admin Login</h3>
    <form>
        Username:<br><input type="text"><br><br>
        Password:<br><input type="password"><br><br>
        <input type="submit" value="Login">
    </form>

    <h3>Add Book</h3>
    <form>
        Book Name:<br><input type="text"><br><br>
        Author:<br><input type="text"><br><br>
        Category:<br><input type="text"><br><br>
        Quantity:<br><input type="number"><br><br>
        <input type="submit" value="Add Book">
    </form>

    <h3>Update / Delete Book</h3>
    <table border="1" cellpadding="8">
        <tr><th>Book ID</th><th>Book Name</th><th>Author</th><th>Action</th></tr>
        <tr><td>101</td><td>Java Programming</td><td>James Gosling</td>
            <td><button type="button">Update</button>
                <button type="button">Delete</button></td></tr>
    </table>

    <h3>Issue Book</h3>
    <form>
        Student ID:<br><input type="text"><br><br>
        Book ID:<br><input type="text"><br><br>
        Issue Date:<br><input type="date"><br><br>
        Return Date:<br><input type="date"><br><br>
        <input type="submit" value="Issue Book">
    </form>

    <h3>Return Book</h3>
    <form>
        Student ID:<br><input type="text"><br><br>
        Book ID:<br><input type="text"><br><br>
        Return Date:<br><input type="date"><br><br>
        <input type="submit" value="Return Book">
    </form>

    <h3>Fine Management</h3>
    <form>
        Student ID:<br><input type="text"><br><br>
        Fine Amount:<br><input type="number"><br><br>
        <input type="submit" value="Save Fine">
    </form>

    <h3>Upload Study Material</h3>
    <form>
        Year:<br>
        <select>
            <option>1st Year</option><option>2nd Year</option>
            <option>3rd Year</option><option>4th Year</option>
        </select><br><br>
        Semester:<br>
        <select>
            <option>Semester 1</option><option>Semester 2</option>
            <option>Semester 3</option><option>Semester 4</option>
            <option>Semester 5</option><option>Semester 6</option>
            <option>Semester 7</option><option>Semester 8</option>
        </select><br><br>
        File:<br><input type="file"><br><br>
```

```
        <input type="submit" value="Upload">
    </form>
</section>
<hr>

<section id="user">
    <h2>User</h2>
    <h3>User Login</h3>
    <form>
        Email:<br><input type="email"><br><br>
        Password:<br><input type="password"><br><br>
        <input type="submit" value="Login">
    </form>

    <h3>Register</h3>
    <form>
        Name:<br><input type="text"><br><br>
        Email:<br><input type="email"><br><br>
        Phone:<br><input type="text"><br><br>
        Password:<br><input type="password"><br><br>
        <input type="submit" value="Register">
    </form>

    <h3>Search Book</h3>
    <input type="text" placeholder="Search Book">
    <button type="button">Search</button>

    <h3>Book Details</h3>
    <table border="1" cellpadding="8">
        <tr><th>Book Name</th><th>Author</th><th>Status</th></tr>
        <tr><td>Java Programming</td><td>James Gosling</td><td>Available</td></tr>
    </table>

    <h3>My Borrow History</h3>
    <table border="1" cellpadding="8">
        <tr><th>Book</th><th>Issue Date</th><th>Expiry Date</th></tr>
        <tr><td>Java Programming</td><td>10-07-2026</td><td>20-07-2026</td></tr>
    </table>

    <h3>Check Fine</h3>
    <p><strong>Total Fine:</strong> ₹100</p>

    <h3>Study Materials</h3>
    <select>
        <option>Select Year</option><option>1st Year</option>
        <option>2nd Year</option><option>3rd Year</option><option>4th Year</option>
    </select>
    <select>
        <option>Select Semester</option><option>Semester 1</option>
        <option>Semester 2</option><option>Semester 3</option><option>Semester 4</option>
        <option>Semester 5</option><option>Semester 6</option><option>Semester 7</option>
        <option>Semester 8</option>
    </select>
    <button type="button">Search</button>
    <p>Database Management System Notes</p>
    <button type="button">Download</button>

    <h3>My Profile</h3>
    <p>Name: Student Name</p>
    <p>Email: student@gmail.com</p>
    <p>Department: Computer Science</p>
</section>
<hr>

<section id="contact">
    <h2>Contact Us</h2>
    <form>
        Name:<br><input type="text"><br><br>
        Email:<br><input type="email"><br><br>
        Message:<br><textarea rows="5" cols="40"></textarea><br><br>
        <input type="submit" value="Send">
    </form>
</section>
<hr>
```

```
        <footer align="center">
            <p>&copy; 2026 Library Management System. All Rights Reserved.</p>
        </footer>
    </body>
    </html>
```

Library Management System

Home | Admin | User | Contact

Welcome

The Library Management System helps students and librarians manage books, issue and return books, check fines, and download study materials efficiently.

Librarian Admin

Admin Login

Username:

Password:

Login

Add Book

Book Name:

Author:

Category:

Quantity:

Add Book

Update / Delete Book

Book ID	Book Name	Author	Action
101	Java Programming	James Gosling	Update Delete

Issue Book

Student ID:

Book ID:

Issue Date:

dd - mm - yyyy

Return Date:

dd - mm - yyyy

Issue Book

Return Book

Student ID:

Book ID:

Return Date:

dd - mm - yyyy

4. SAMPLE INPUT

The webpage displays sample information in the browser. The following values can be entered manually in the form controls to demonstrate the interface.

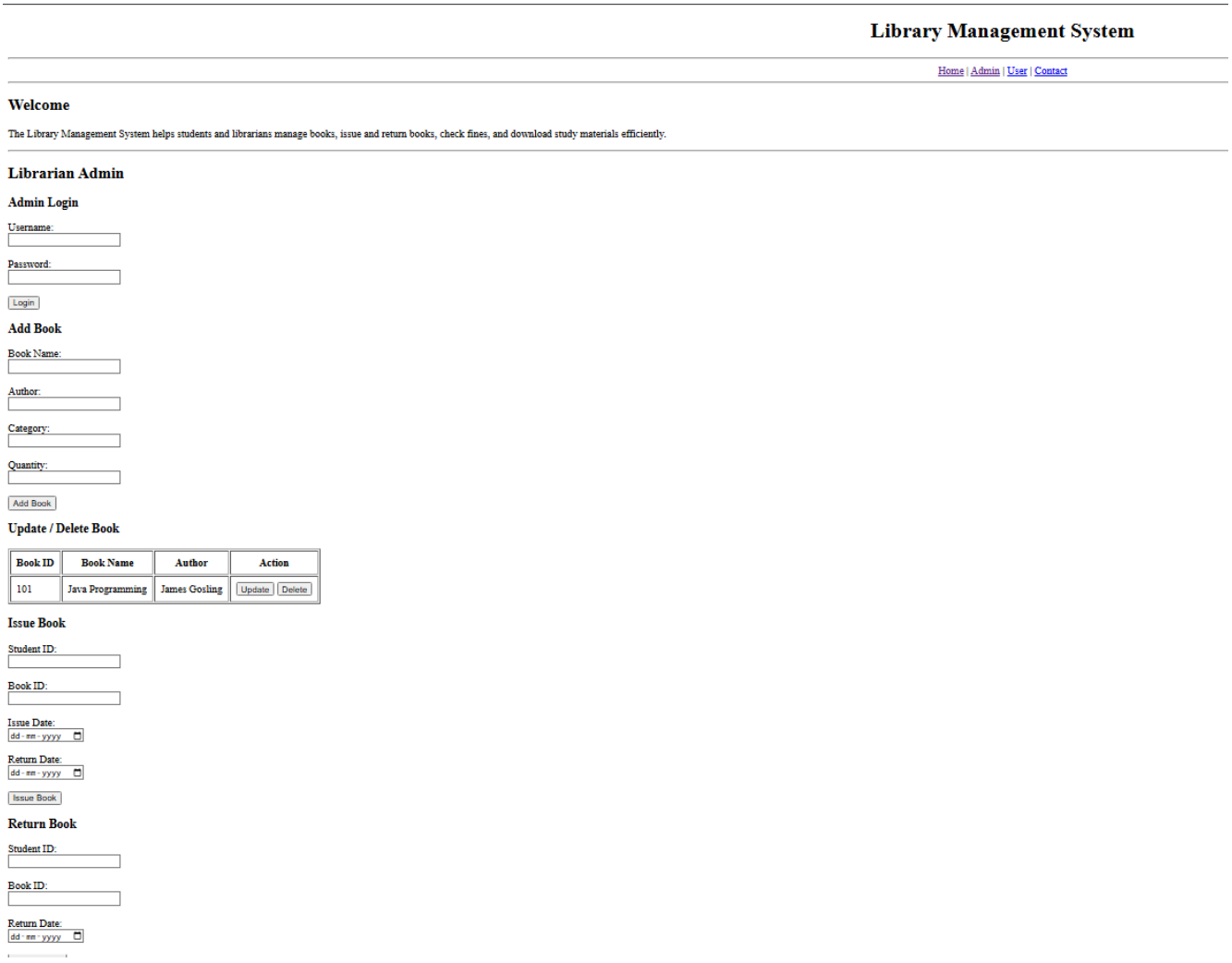
Field / Feature	Sample Input
Admin username	librarian01
Admin password	admin123 (demonstration only)
Book name	Python Programming
Author	Guido van Rossum
Category	Programming
Quantity	5
Student ID for issue	STU2026
Book ID	102
Issue date	2026-10-23
Return date	2026-10-30
Fine amount	100
User name	Student Name
User email	student@gmail.com
Search text	Java Programming
Study material selection	2nd Year, Semester 3
Contact message	Please provide information about available books.

5. SAMPLE OUTPUT

When library_management.html is opened in a browser, the page displays the following sections and sample records:

- Page heading: Library Management System.
- Navigation menu: Home | Admin | User | Contact.
- Welcome section describing the purpose of the system.
- Admin section containing login, Add Book, Update/Delete Book, Issue Book, Return Book, Fine Management, and Upload Study Material forms.
- Sample admin book table: Book ID 101 | Java Programming | James Gosling | Update / Delete.
- User section containing login, registration, search controls, book details, borrowing history, fine amount, study materials, and profile information.
- Sample book availability: Java Programming by James Gosling — Available.
- Sample borrowing history: Java Programming — Issue Date 10-07-2026 — Expiry Date 20-07-2026.
- Sample fine display: Total Fine: ₹100.
- Contact form and footer copyright notice.
-
-

Result:



Library Management System

Home | Admin | User | Contact

Welcome

The Library Management System helps students and librarians manage books, issue and return books, check fines, and download study materials efficiently.

Librarian Admin

Admin Login

Username:

Password:

Login

Add Book

Book Name:

Author:

Category:

Quantity:

Add Book

Update / Delete Book

Book ID	Book Name	Author	Action
101	Java Programming	James Gosling	Update Delete

Issue Book

Student ID:

Book ID:

Issue Date:

dd - mm - yyyy

Return Date:

dd - mm - yyyy

Issue Book

Return Book

Student ID:

Book ID:

Return Date:

dd - mm - yyyy

HOTEL ROOM BOOKING SYSTEM

1. AIM

To design and develop a Hotel Room Booking System webpage using HTML and CSS. The webpage displays room types and prices, provides a booking form for guest details, check-in and check-out dates, number of guests and room type, and presents hotel information and contact details.

2. PROCEDURE

1. Open a text editor such as Notepad or Visual Studio Code.
2. Create a new file and save it as hotel_room_booking.html.
3. Write the basic HTML structure using <!DOCTYPE html>, <html>, <head>, and <body> tags.
4. Add CSS inside the <style> element to style the page background, header, navigation bar, room cards, booking form, information sections, and footer.
5. Create a header with the title “Hotel Room Booking System” and a welcome message.
6. Create navigation links and a hero section introducing the booking service.
7. Display four room cards: Single Room, Double Room, Deluxe Room, and Suite Room, with their prices and facilities.
8. Create a booking form with name, email, check-in date, check-out date, number of guests, and room type fields.
9. Add About Us and Contact Us sections with hotel service and contact information.
10. Add a footer containing the copyright notice.
11. Save the file and open it in a web browser to view the webpage.
12. Enter sample details and test required-field validation. Actual booking storage and confirmation require JavaScript and/or a server-side application with a database.

3. PROGRAM CODE

Save the following code as hotel_room_booking.html.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Hotel Room Booking</title>
<style>
*{margin:0;padding:0;box-sizing:border-box;font-family:Arial,sans-serif;}
body{background:#f4f4f4;}
header{background:#003366;color:white;padding:20px;text-align:center;}
nav{background:#0059b3;padding:10px;text-align:center;}
nav a{color:white;text-decoration:none;margin:15px;font-weight:bold;}
.hero{text-align:center;background:#e6f2ff;padding:40px;}
.hero h2{margin-bottom:10px;}
.container{width:90%;margin:auto;padding:20px;}
.rooms{display:flex;justify-content:space-around;flex-wrap:wrap;}
.room{background:white;width:250px;margin:10px;padding:15px;border-radius:10px;
box-shadow:0 0 10px gray;text-align:center;}
.room h3{color:#003366;}
.room button{background:#28a745;color:white;border:none;padding:10px;margin-top:10px;
cursor:pointer;border-radius:5px;}
form{background:white;padding:20px;margin-top:20px;border-radius:10px;box-shadow:0 0 10px gray;}
input,select{width:100%;padding:10px;margin:10px 0;}
input[type=submit]{background:#003366;color:white;border:none;cursor:pointer;}
.about,.contact{background:white;margin-top:20px;padding:20px;border-radius:10px;
box-shadow:0 0 10px gray;}
footer{background:#003366;color:white;text-align:center;padding:15px;margin-top:20px;}
</style>
</head>
<body>
<header>
<h1>Hotel Room Booking System</h1>
```

```
<p>Welcome to Our Hotel</p>
</header>
<nav>
<a href="#">Home</a> <a href="#">Rooms</a> <a href="#">Booking</a>
<a href="#">About</a> <a href="#">Contact</a>
</nav>
<section class="hero">
<h2>Book Your Dream Room</h2>
<p>Easy, Fast and Secure Booking</p>
</section>
<div class="container">
<h2>Room Types</h2>
<div class="rooms">
<div class="room"><h3>Single Room</h3><p>Price: ₹1500 / Night</p>
<p>1 Bed, Free WiFi</p><button type="button">Book Now</button></div>
<div class="room"><h3>Double Room</h3><p>Price: ₹2500 / Night</p>
<p>2 Beds, AC, TV</p><button type="button">Book Now</button></div>
<div class="room"><h3>Deluxe Room</h3><p>Price: ₹4000 / Night</p>
<p>Luxury Room with Breakfast</p><button type="button">Book Now</button></div>
<div class="room"><h3>Suite Room</h3><p>Price: ₹6000 / Night</p>
<p>Premium Luxury Suite</p><button type="button">Book Now</button></div>
</div>
<h2>Booking Form</h2>
<form>
<label>Name</label><input type="text" placeholder="Enter Your Name" required>
<label>Email</label><input type="email" placeholder="Enter Email" required>
<label>Check-in Date</label><input type="date" required>
<label>Check-out Date</label><input type="date" required>
<label>Guests</label><select><option>1</option><option>2</option><option>3</option><option>4</option></select>
<label>Room Type</label><select><option>Single Room</option><option>Double Room</option>
<option>Deluxe Room</option><option>Suite Room</option></select>
<input type="submit" value="Book Room">
</form>
<div class="about"><h2>About Us</h2>
<p>Our hotel provides comfortable rooms with modern facilities. We offer secure booking,
friendly service, free WiFi, 24/7 customer support, and affordable prices.</p></div>
<div class="contact"><h2>Contact Us</h2>
<p>Email: hotel@example.com</p><p>Phone: +91 9876543210</p>
<p>Address: Chennai, Tamil Nadu</p></div>
</div>
<footer><p>© 2026 Hotel Room Booking System | Internet Programming Lab</p></footer>
</body>
</html>
```

4. SAMPLE INPUT

The following sample values can be entered into the booking form.

Field	Sample Input
Name	Arun Kumar
Email	arun.kumar@example.com
Check-in Date	2026-11-05
Check-out Date	2026-11-08
Guests	2
Room Type	Double Room

5. SAMPLE OUTPUT

The following screenshot is an illustrative output preview of the Hotel Room Booking System webpage. The exact appearance may vary slightly by browser and screen size.

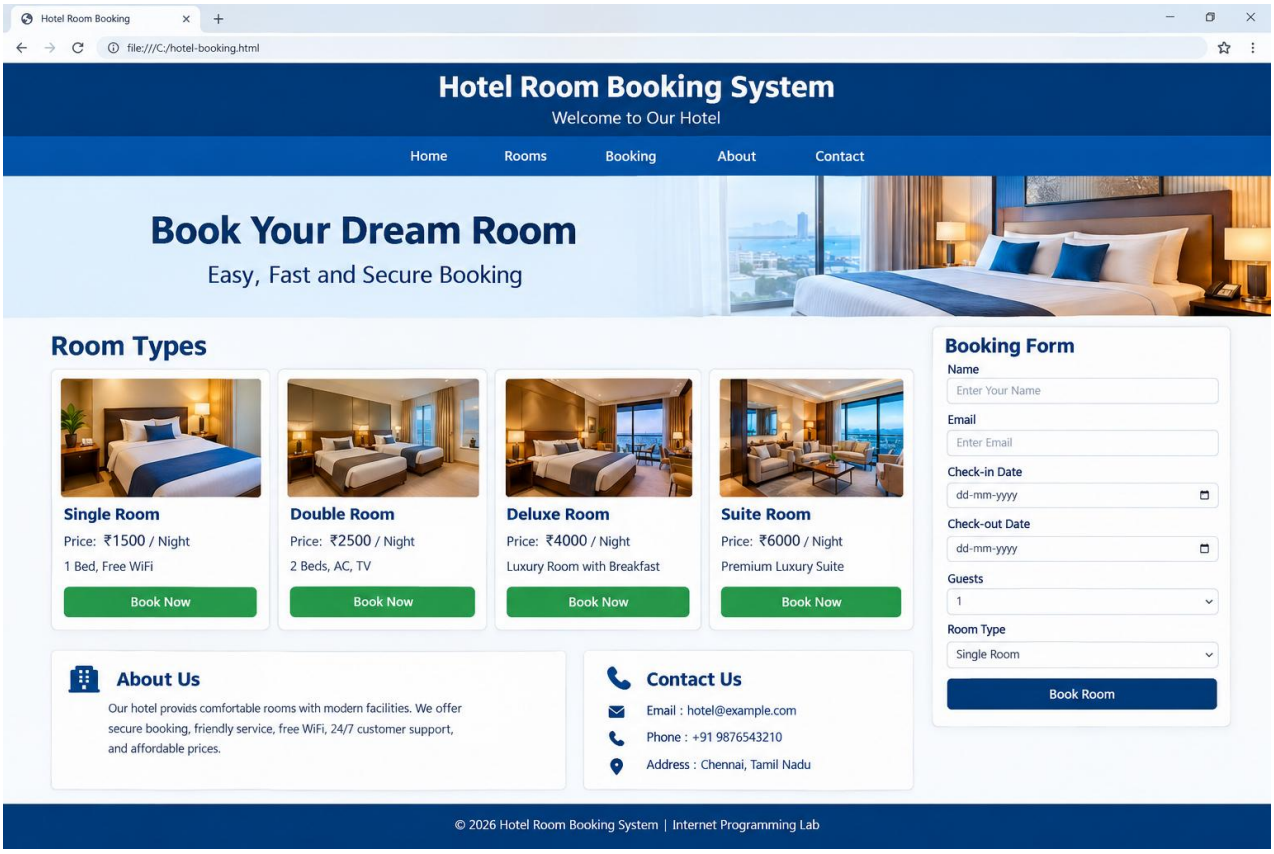


Figure 1: Sample output of the Hotel Room Booking System

Expected output details:

- A dark-blue header displays the system title and welcome message.
- A blue navigation bar displays Home, Rooms, Booking, About, and Contact.
- A hero section displays “Book Your Dream Room” and “Easy, Fast and Secure Booking”.
- Four room cards display Single Room (₹1500/night), Double Room (₹2500/night), Deluxe Room (₹4000/night), and Suite Room (₹6000/night).
- Each room card includes room facilities and a Book Now button.
- The booking form displays name, email, check-in date, check-out date, guests, and room type fields.

- About Us and Contact Us sections show hotel information and contact details.
- The footer displays the Internet Programming Lab copyright text.

6. RESULT

Thus, the Hotel Room Booking System webpage was designed using HTML and CSS. The webpage presents room options, prices, a booking form, hotel information, and contact details in an organized layout. The interface can be enhanced with JavaScript and backend/database integration to validate dates, calculate booking costs, check room availability, save reservations, and generate booking confirmations.

ONLINE BUS TICKET BOOKING SYSTEM

1. AIM

To design and develop an Online Bus Ticket Booking System webpage using HTML and CSS. The webpage enables users to enter journey details, view sample available buses and fares, fill in passenger and seat details, and view user features, admin features, and a comments form.

2. PROCEDURE

1. Open a text editor such as Notepad or Visual Studio Code.
2. Create a new file and save it as bus_booking.html.
3. Write the basic HTML structure using <!DOCTYPE html>, <html>, <head>, and <body> tags.
4. Add the page title and CSS styles inside the <style> element.
5. Create a header and navigation bar for Home, Search Bus, Book Ticket, Login, Register, and Contact.
6. Create the Search Bus form with From, To, date, and time selection fields.
7. Display available buses in a table with bus name, category, seats, and fare.
8. Create the Book Ticket form with passenger name, email, mobile number, bus selection, and seat selection.
9. Add lists describing user features and admin features.
10. Create a Post Comments form with name and comment fields.
11. Add a footer with the copyright notice and lab experiment title.
12. Save the file and open it in a web browser to view the page. Real-time bus search, booking, seat locking, payments, and database storage require JavaScript and backend integration.

3. PROGRAM CODE

Save the following complete program as bus_booking.html.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Online Bus Ticket Booking System</title>
<style>
*{margin:0;padding:0;box-sizing:border-box;font-family:Arial,sans-serif;}
body{background:#f4f4f4;}
header{background:#004080;color:white;text-align:center;padding:20px;}
nav{background:#0066cc;padding:12px;text-align:center;}
nav a{color:white;text-decoration:none;margin:15px;font-weight:bold;}
.container{width:90%;margin:auto;padding:20px;}
.section{background:white;padding:20px;margin-top:20px;border-radius:8px;box-shadow:0 0 8px gray;}
h2{color:#004080;margin-bottom:10px;}
input,select,textarea{width:100%;padding:10px;margin:8px 0;}
button{background:#004080;color:white;padding:10px 20px;border:none;cursor:pointer;}
table{width:100%;border-collapse:collapse;margin-top:10px;}
table,th,td{border:1px solid gray;}
th{background:#004080;color:white;}
th,td{padding:10px;text-align:center;}
footer{background:#004080;color:white;text-align:center;padding:15px;margin-top:20px;}
</style>
</head>
<body>
<header>
<h1>Online Bus Ticket Booking System</h1>
<p>Book Your Bus Tickets Easily</p>
</header>
<nav>
<a href="#">Home</a><a href="#">Search Bus</a><a href="#">Book Ticket</a>
<a href="#">Login</a><a href="#">Register</a><a href="#">Contact</a>
</nav>
<div class="container">
<div class="section">
<h2>Search Bus</h2>
<form>
```

```
<input type="text" placeholder="From">
<input type="text" placeholder="To">
<input type="date">
<select><option>Select Time</option><option>Morning</option>
<option>Afternoon</option><option>Evening</option><option>Night</option></select>
<button type="submit">Search Bus</button>
</form>
</div>
<div class="section">
<h2>Available Buses</h2>
<table>
<tr><th>Bus Name</th><th>Category</th><th>Seats</th><th>Fare</th></tr>
<tr><td>Express Travels</td><td>Volvo</td><td>25</td><td>₹900</td></tr>
<tr><td>Royal Bus</td><td>Sleeper</td><td>18</td><td>₹1200</td></tr>
<tr><td>City Coach</td><td>AC Coach</td><td>30</td><td>₹700</td></tr>
</table>
</div>
<div class="section">
<h2>Book Ticket</h2>
<form>
<input type="text" placeholder="Passenger Name">
<input type="email" placeholder="Email">
<input type="tel" placeholder="Mobile Number">
<select><option>Select Bus</option><option>Express Travels</option>
<option>Royal Bus</option><option>City Coach</option></select>
<select><option>Select Seat</option><option>A1</option><option>A2</option>
<option>B1</option><option>B2</option></select>
<button type="button">Book Ticket</button>
</form>
</div>
<div class="section">
<h2>User Features</h2>
<ul>
<li>User Registration</li><li>User Login</li><li>Search Bus</li>
<li>Book Ticket</li><li>Cancel Ticket</li><li>Booking History</li>
<li>Payment Gateway</li><li>Email Confirmation</li>
</ul>
</div>
<div class="section">
<h2>Admin Features</h2>
<ul>
<li>Manage Bus Categories</li><li>Add / Update Bus Information</li>
<li>Manage Seat Availability</li><li>Manage Registered Users</li>
<li>Generate Reports</li><li>Manage Comments</li><li>Manage News</li>
</ul>
</div>
<div class="section">
<h2>Post Comments</h2>
<form>
<input type="text" placeholder="Your Name">
<textarea rows="4" placeholder="Write Your Comment"></textarea>
<button type="button">Post Comment</button>
</form>
</div>
</div>
<footer>
<p>© 2026 Online Bus Ticket Booking System</p>
<p>Internet Programming Lab Experiment</p>
</footer>
</body>
</html>
```

4. SAMPLE INPUT

The following sample values can be entered into the search and ticket booking forms.

Field	Sample Input
From	Chennai
To	Bangalore
Journey Date	2026-11-05
Time	Evening
Passenger Name	Arun Kumar
Email	arun.kumar@example.com
Mobile Number	9876543210
Bus	Express Travels
Seat	A1
Comment	The booking page is easy to understand.

5. SAMPLE OUTPUT

The screenshot below is an illustrative output preview of the Online Bus Ticket Booking System webpage. The actual page appearance may vary slightly by browser and screen size.

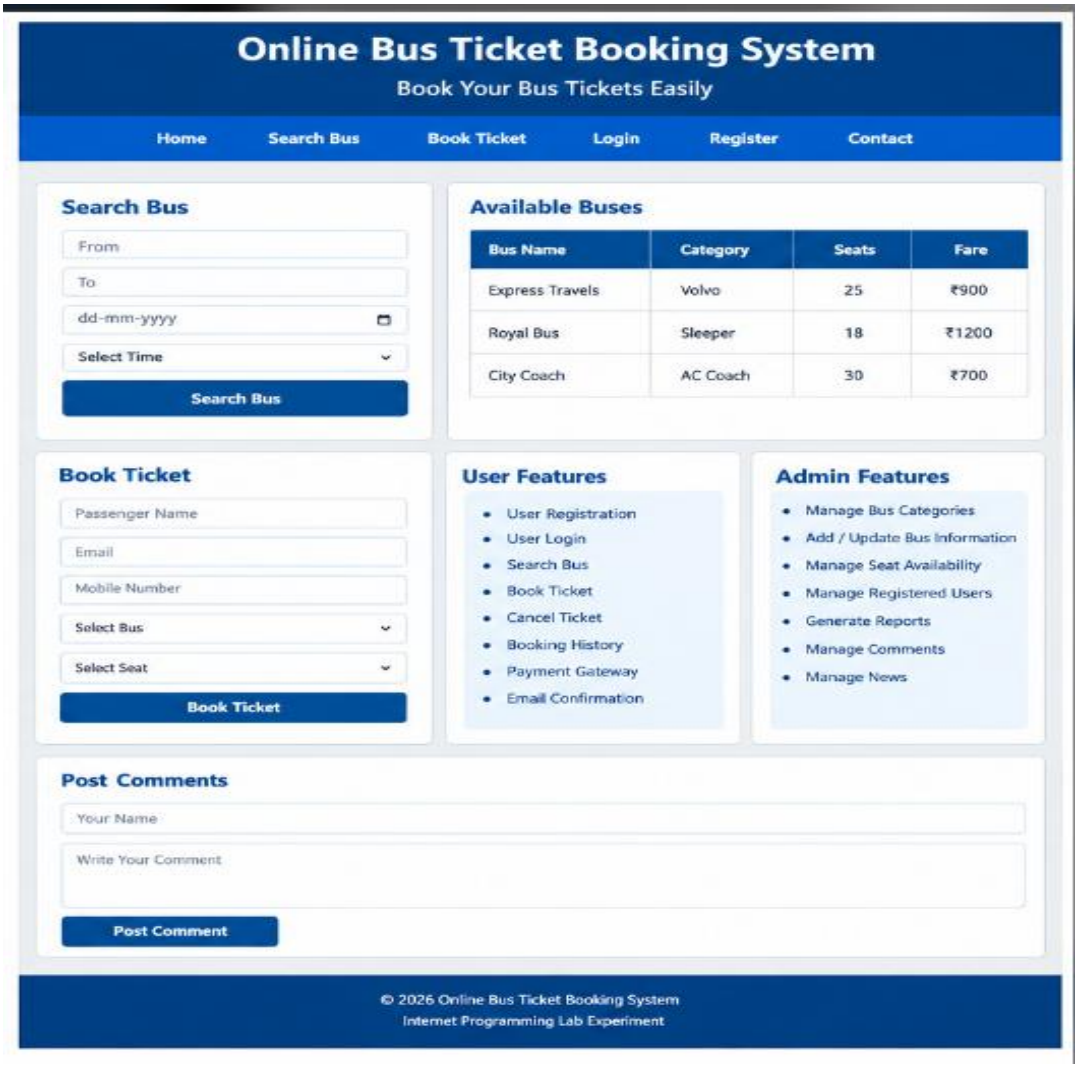


Figure 1: Sample output of the Online Bus Ticket Booking System

Expected output details:

- Header: Online Bus Ticket Booking System — Book Your Bus Tickets Easily.
- Navigation menu: Home, Search Bus, Book Ticket, Login, Register, and Contact.
- Search Bus section with From, To, journey date, and time fields.
- Available Buses table with Express Travels (Volvo, 25 seats, ₹900), Royal Bus (Sleeper, 18 seats, ₹1200), and City Coach (AC Coach, 30 seats, ₹700).
- Book Ticket form with passenger name, email, mobile number, bus selection, and seat selection.
- User Features list and Admin Features list.
- Post Comments form with name and comment fields.
- Footer showing copyright and Internet Programming Lab Experiment.

6. RESULT

Thus, the Online Bus Ticket Booking System webpage was designed using HTML and CSS. It presents bus search fields, sample bus availability and fares, ticket booking inputs, user and admin feature lists, and a comments section. The page is a front-end demonstration; actual booking confirmation, seat availability updates, cancellation, payment processing, login, and data storage require JavaScript and backend/database integration.

ONLINE TRAIN RESERVATION SYSTEM

1. AIM

To design and develop a simple Online Train Reservation System webpage using HTML and CSS that presents train-search fields, sample train availability, ticket-booking fields, PNR-status input, user and administrator features, and a refund-policy section.

2. PROCEDURE

1. Open a text editor or an IDE such as Visual Studio Code.
2. Create a new file and save it as index.html.
3. Write the basic HTML document structure with the DOCTYPE, head, title, and body elements.
4. Add CSS styles to format the page, including the header, navigation bar, content sections, forms, table, buttons, and footer.
5. Create the Search Train form with origin, destination, journey date, and class fields.
6. Display sample available trains in a table with train number, train name, departure, arrival, and seat count.
7. Create the Book Ticket and Check PNR Status forms.
8. Add lists describing user features and administrator features, followed by the refund policy.
9. Save the file and open index.html in a web browser to view the webpage.
10. Verify that the page layout and sample data are displayed correctly. The supplied code is a front-end design; the forms do not perform real booking or PNR lookup without JavaScript and a backend.

3. PROGRAM CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Online Train Reservation System</title>
<style>
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: Arial, sans-serif;
}
body { background: #f2f2f2; }
header {
    background: #003366;
    color: white;
    text-align: center;
    padding: 20px;
}
nav {
    background: #0059b3;
    padding: 12px;
    text-align: center;
}
nav a {
    color: white;
    text-decoration: none;
    margin: 15px;
    font-weight: bold;
}
.container {
    width: 90%;
    margin: auto;
    padding: 20px;
}
.section {
    background: white;
```

```
    padding: 20px;
    margin-top: 20px;
    border-radius: 8px;
    box-shadow: 0 0 8px gray;
}
h2 { color: #003366; margin-bottom: 10px; }
input, select {
    width: 100%;
    padding: 10px;
    margin: 8px 0;
}
button {
    background: #003366;
    color: white;
    border: none;
    padding: 10px 20px;
    cursor: pointer;
}
table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 10px;
}
table, th, td { border: 1px solid #999; }
th { background: #003366; color: white; }
th, td { padding: 10px; text-align: center; }
footer {
    background: #003366;
    color: white;
    text-align: center;
    padding: 15px;
    margin-top: 20px;
}
ul { margin-left: 20px; }
</style>
</head>
<body>
<header>
    <h1>Online Train Reservation System</h1>
    <p>Book Train Tickets Easily</p>
</header>
<nav>
    <a href="#">Home</a>
    <a href="#">Search Train</a>
    <a href="#">Book Ticket</a>
    <a href="#">PNR Status</a>
    <a href="#">Login</a>
    <a href="#">Register</a>
</nav>
<div class="container">
    <div class="section">
        <h2>Search Train</h2>
        <form>
            <input type="text" placeholder="From Station">
            <input type="text" placeholder="To Station">
            <input type="date">
            <select>
                <option>Select Class</option>
                <option>Sleeper</option><option>3A</option>
                <option>2A</option><option>1A</option>
            </select>
            <button>Search Train</button>
        </form>
    </div>
    <div class="section">
        <h2>Available Trains</h2>
        <table>
            <tr><th>Train No</th><th>Train Name</th><th>Departure</th>
                <th>Arrival</th><th>Seats</th></tr>
            <tr><td>12623</td><td>Chennai Express</td><td>08:00 AM</td>
                <td>04:00 PM</td><td>120</td></tr>
            <tr><td>12645</td><td>Super Fast Express</td><td>09:30 AM</td>
                <td>06:30 PM</td><td>75</td></tr>
            <tr><td>12789</td><td>Intercity Express</td><td>11:00 AM</td>
                <td>08:00 PM</td><td>90</td></tr>
        </table>
    </div>
    <div class="section">
        <h2>Book Ticket</h2>
        <form>
            <input type="text" placeholder="Passenger Name">
            <input type="number" placeholder="Age">
            <select>
                <option>Gender</option><option>Male</option>
                <option>Female</option><option>Other</option>
            </select>
            <input type="text" placeholder="Origin Station">
            <input type="text" placeholder="Destination Station">
```

```
            <select>
                <option>Select Class</option><option>Sleeper</option>
                <option>3A</option><option>2A</option><option>1A</option>
            </select>
            <button>Book Ticket</button>
        </form>
    </div>
    <div class="section">
        <h2>Check PNR Status</h2>
        <form>
            <input type="text" placeholder="Enter PNR Number">
            <button>Check Status</button>
        </form>
    </div>
    <div class="section">
        <h2>User Features</h2>
        <ul>
            <li>User Registration</li><li>User Login</li>
            <li>Search Trains</li><li>Seat Availability</li>
            <li>Book Tickets</li><li>View Booking History</li>
            <li>Print Ticket</li><li>Cancel Ticket</li>
            <li>Online/Offline Payment</li>
        </ul>
    </div>
    <div class="section">
        <h2>Admin Features</h2>
        <ul>
            <li>Manage Train Details</li><li>Manage Train Schedule</li>
            <li>Manage Seat Availability</li><li>Manage User Accounts</li>
            <li>Manage Fare Details</li><li>Manage PNR Details</li>
            <li>Manage Refund Requests</li><li>Generate Reports</li>
        </ul>
    </div>
    <div class="section">
        <h2>Refund Policy</h2>
        <p>Passengers can cancel tickets using the PNR number.
        Refunds are processed based on railway cancellation rules and the
        cancellation date.</p>
    </div>
</div>
<footer>
    <p>© 2026 Online Train Reservation System</p>
    <p>Internet Programming Lab Experiment</p>
</footer>
</body>
</html>
```

4. INPUT

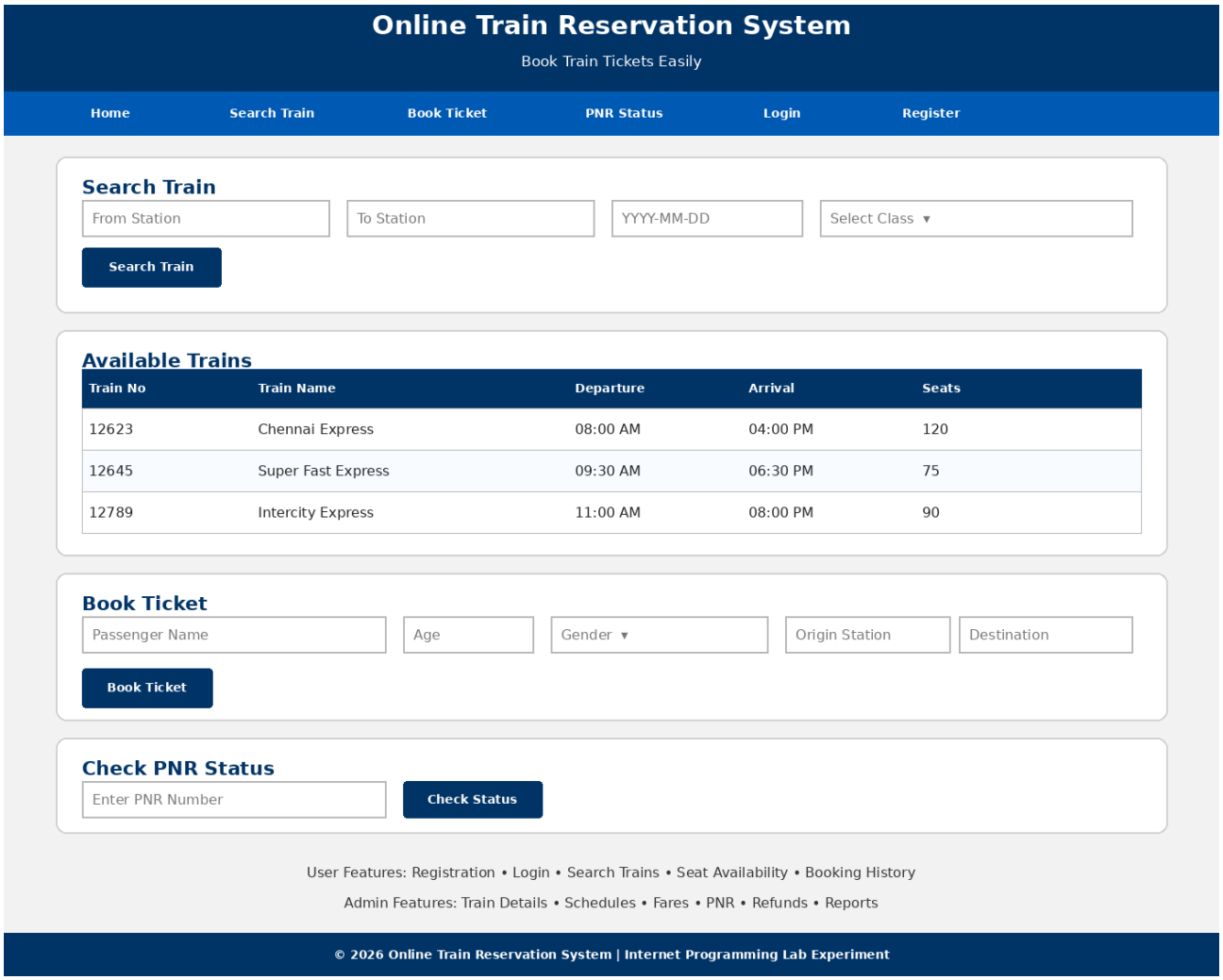
The webpage accepts the following user-entered values through its form fields:

- Search Train: From Station, To Station, Journey Date, and Class (Sleeper, 3A, 2A, or 1A).
- Book Ticket: Passenger Name, Age, Gender, Origin Station, Destination Station, and Class.
- Check PNR Status: PNR Number.

Note: The provided HTML does not validate or process these values because no JavaScript or server-side code is included.

5. OUTPUT

The browser displays the Online Train Reservation System interface with a header and navigation bar, train-search form, available-trains table, ticket-booking form, PNR-status form, user/admin feature lists, refund policy, and footer.



6. RESULT

Thus, the front-end design of the Online Train Reservation System was created successfully using HTML and CSS. The page presents the main sections required for searching trains, viewing sample train details, entering passenger booking information, checking PNR status, and displaying user/admin features. Actual

ticket booking, PNR verification, payment, and refund processing require JavaScript and a backend/database and are not implemented in the supplied code.

EXP-5 ONLINE FLIGHT RESERVATION SYSTEM

1. AIM

To design and develop an Online Flight Reservation System using HTML, CSS, and JavaScript. The application provides a front-end interface for user registration and login, flight search, ticket booking, payment simulation, ticket viewing and cancellation, feedback submission, and basic administrator operations.

2. PROCEDURE

1. Open a text editor or IDE such as Visual Studio Code.
2. Create a new file and save the complete program as index.html.
3. Write the HTML structure to define the header, navigation menu, content pages, forms, tables, and footer.
4. Use CSS to style the page layout, cards, form controls, buttons, flight results, tables, and responsive mobile view.
5. Add JavaScript functions to switch between pages, register and log in a customer, reset a password message, and search the sample flight list.
6. Implement the booking flow by selecting a flight, entering passenger details, selecting a seat, and proceeding to the payment screen.
7. Implement the payment simulation to generate a booking ID, display ticket details, update the booking count and revenue, and reduce the selected flight's seat count.
8. Add ticket printing and cancellation, booking-status checking, feedback submission, administrator login, flight entry, and report features.
9. Save the file and open index.html in a web browser.
10. Test the pages and functions using the sample data provided in the program.

3. PROGRAM CODE

Complete HTML, CSS, and JavaScript code from the supplied source is included below. For readability, the code is formatted in a monospaced font.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Online Flight Reservation System</title>


<style>
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial,sans-serif;
}


body{
    background:#f2f6fa;
    color:#222;
}


header{
    background:#073b66;
    color:white;
    padding:18px 40px;
    display:flex;
    justify-content:space-between;
    align-items:center;
}


header h1{
    font-size:24px;
```

```
}

nav button{
    background:none;
    border:none;
    color:white;
    margin:5px;
    cursor:pointer;
    font-size:15px;
}

.container{
    width:90%;
    max-width:1100px;
    margin:30px auto;
}

.page{
    display:none;
}

.active{
    display:block;
}

.card{
    background:white;
    padding:25px;
    border-radius:12px;
    box-shadow:0 3px 12px #ccc;
    margin-bottom:20px;
}

h2{
    color:#073b66;
    margin-bottom:20px;
}

input,select{
    width:100%;
    padding:12px;
    margin:8px 0 15px;
    border:1px solid #bbb;
    border-radius:6px;
}

button{
    background:#0878c9;
    color:white;
    border:none;
    padding:11px 18px;
    border-radius:6px;
    cursor:pointer;
}

button:hover{
    background:#055c99;
}

.search-grid{
    display:grid;
    grid-template-columns:repeat(4,1fr);
    gap:15px;
}

.flight{
    border:1px solid #ddd;
    padding:18px;
    border-radius:10px;
    margin:12px 0;
    background:#fafafa;
}

.flight button{
    float:right;
}

table{
    width:100%;
    border-collapse:collapse;
    background:white;
}

th,td{
    padding:12px;
    border:1px solid #ddd;
```

```
    text-align:center;
}

th{
    background:#073b66;
    color:white;
}

.message{
    padding:12px;
    margin:10px 0;
    background:#e7f5ff;
    border-radius:6px;
}

.danger{
    background:#d9534f;
}

footer{
    text-align:center;
    background:#073b66;
    color:white;
    padding:20px;
    margin-top:40px;
}

@media(max-width:700px){
    .search-grid{
        grid-template-columns:1fr;
    }

    header{
        flex-direction:column;
        gap:10px;
        text-align:center;
    }

    table{
        font-size:12px;
    }
}
</style>
</head>

<body>

<header>
    <h1>✈ SkyFly Reservation</h1>

    <nav>
        <button onclick="showPage('home')">Home</button>
        <button onclick="showPage('search')">Search Flight</button>
        <button onclick="showPage('myticket')">My Ticket</button>
        <button onclick="showPage('feedback')">Feedback</button>
        <button onclick="showPage('admin')">Admin</button>
        <button onclick="showPage('login')">Login</button>
    </nav>
</header>

<div class="container">

<!-- HOME -->
<section id="home" class="page active">

    <div class="card">
        <h2>Online Flight Reservation System</h2>
        <p>
            Search flights, check real-time seat availability,
            book tickets, make payments and manage your reservations
            easily from one platform.
        </p>
        <br>
        <button onclick="showPage('search')">Search Flights</button>
    </div>

    <div class="card">
        <h2>Features</h2>
        <p>✓ Flight Search</p>
        <p>✓ Seat Selection</p>
        <p>✓ Online Booking</p>
        <p>✓ Secure Payment</p>
        <p>✓ E-Ticket</p>
        <p>✓ Booking History</p>
```

```
        <p>✓ Ticket Cancellation</p>
        <p>✓ Customer Feedback</p>
        <p>✓ Admin Reports</p>
    </div>

</section>



<!-- LOGIN -->
<section id="login" class="page">

<div class="card">
<h2>Customer Login</h2>

<input type="email" id="loginEmail" placeholder="Email">
<input type="password" id="loginPassword" placeholder="Password">

<button onclick="login()">Login</button>
<button onclick="showPage('register')">Register</button>
<button onclick="showPage('forgot')">Forgot Password?</button>

<p id="loginMsg"></p>
</div>

</section>



<!-- REGISTER -->
<section id="register" class="page">

<div class="card">
<h2>Create Account</h2>

<input id="regName" placeholder="Full Name">
<input id="regEmail" type="email" placeholder="Email">
<input id="regPassword" type="password" placeholder="Password">

<button onclick="register()">Register</button>
<button onclick="showPage('login')">Back to Login</button>

<p id="registerMsg"></p>
</div>

</section>



<!-- FORGOT PASSWORD -->
<section id="forgot" class="page">

<div class="card">
<h2>Forgot Password</h2>

<input type="email" id="forgotEmail" placeholder="Enter registered email">

<button onclick="resetPassword()">Reset Password</button>

<p id="forgotMsg"></p>
</div>

</section>



<!-- SEARCH FLIGHT -->
<section id="search" class="page">

<div class="card">
<h2>Search Flights</h2>

<div class="search-grid">

<div>
<label>From</label>
<select id="from">
<option>Chennai</option>
<option>Bangalore</option>
<option>Delhi</option>
<option>Mumbai</option>
```

```
</select>
</div>


<div>
<label>Destination</label>
<select id="to">
<option>Delhi</option>
<option>Mumbai</option>
<option>Bangalore</option>
<option>Chennai</option>
</select>
</div>


<div>
<label>Date</label>
<input type="date" id="flightDate">
</div>


<div>
<label>Class</label>
<select id="flightClass">
<option>Economy</option>
<option>Business</option>
<option>First Class</option>
</select>
</div>


</div>


<button onclick="searchFlights()">Search Flights</button>


</div>


<div id="flightResults"></div>


</section>



<!-- BOOK TICKET -->
<section id="booking" class="page">


<div class="card">


<h2>Book Ticket</h2>


<div id="selectedFlight"></div>


<h3>Passenger Details</h3>


<input id="passengerName" placeholder="Passenger Name">
<input id="passengerAge" type="number" placeholder="Age">
<input id="passengerPassport" placeholder="Passport / ID Number">


<h3>Select Seat</h3>


<select id="seat">
<option>1A</option>
<option>1B</option>
<option>1C</option>
<option>2A</option>
<option>2B</option>
<option>2C</option>
<option>3A</option>
<option>3B</option>
</select>


<br><br>


<button onclick="proceedPayment()">Continue to Payment</button>


</div>


</section>



<!-- PAYMENT -->
<section id="payment" class="page">


<div class="card">
```

```
<h2>Ticket Payment</h2>

<h3 id="paymentAmount"></h3>

<select id="paymentMethod">
<option>Credit Card</option>
<option>Debit Card</option>
<option>UPI</option>
<option>Net Banking</option>
</select>

<input placeholder="Card / UPI Number">

<input type="password" placeholder="PIN / CVV">

<button onclick="makePayment()">Pay Now</button>

<p id="paymentMsg"></p>

</div>

</section>



<!-- MY TICKET -->
<section id="myticket" class="page">

<div class="card">

<h2>My Ticket</h2>

<div id="ticketDetails">
<p>No ticket booked yet.</p>
</div>

<br>

<button onclick="printTicket()">Print Ticket</button>
<button class="danger" onclick="cancelTicket()">Cancel Ticket</button>

</div>

</section>



<!-- BOOKING STATUS -->
<section id="status" class="page">

<div class="card">

<h2>View Booking Status</h2>

<input id="bookingId" placeholder="Enter Booking ID">

<button onclick="checkStatus()">Check Status</button>

<p id="statusResult"></p>

</div>

</section>



<!-- FEEDBACK -->
<section id="feedback" class="page">

<div class="card">

<h2>Customer Feedback</h2>

<input id="feedbackName" placeholder="Your Name">

<select id="rating">
```

```
<option>5 - Excellent</option>
<option>4 - Very Good</option>
<option>3 - Good</option>
<option>2 - Average</option>
<option>1 - Poor</option>
</select>


<textarea id="feedbackText"
placeholder="Write your feedback"
style="width:100%;height:120px;padding:12px;"></textarea>


<br><br>


<button onclick="submitFeedback()">Submit Feedback</button>


<p id="feedbackMsg"></p>


</div>


</section>



<!-- ADMIN LOGIN -->
<section id="admin" class="page">


<div class="card">


<h2>Administrator Login</h2>


<input id="adminUser" placeholder="Admin Username">


<input id="adminPass" type="password" placeholder="Password">


<button onclick="adminLogin()">Admin Login</button>


<p id="adminMsg"></p>


</div>


</section>



<!-- ADMIN PANEL -->
<section id="adminPanel" class="page">


<div class="card">


<h2>Admin Panel</h2>


<button onclick="showPage('adminFlights')">Manage Flights</button>
<button onclick="showPage('adminUsers')">Manage Users</button>
<button onclick="showPage('report')">Admin Report</button>


</div>


</section>



<!-- ADMIN FLIGHTS -->
<section id="adminFlights" class="page">


<div class="card">


<h2>Manage Flight Schedules</h2>


<input id="newFlight" placeholder="Flight Number">


<input id="newRoute" placeholder="Route">


<input id="newSeats" type="number" placeholder="Number of Seats">


<input id="newFare" type="number" placeholder="Fare">


<button onclick="addFlight()">Add Flight</button>
```

```
<br><br>

<table>

<thead>
<tr>
<th>Flight</th>
<th>Route</th>
<th>Seats</th>
<th>Fare</th>
</tr>
</thead>

<tbody id="flightTable"></tbody>

</table>

</div>

</section>



<!-- ADMIN USERS -->
<section id="adminUsers" class="page">

<div class="card">

<h2>User Accounts</h2>

<table>

<tr>
<th>Name</th>
<th>Email</th>
<th>Status</th>
</tr>

<tr>
<td>Arun</td>
<td>arun@gmail.com</td>
<td>Active</td>
</tr>

<tr>
<td>Priya</td>
<td>priya@gmail.com</td>
<td>Active</td>
</tr>

</table>

</div>

</section>



<!-- REPORT -->
<section id="report" class="page">

<div class="card">

<h2>Admin Report</h2>

<table>

<tr>
<th>Total Flights</th>
<th>Total Bookings</th>
<th>Available Seats</th>
<th>Revenue</th>
</tr>

<tr>
<td id="totalFlights">5</td>
<td id="totalBookings">0</td>
<td>245</td>
<td id="revenue">₹0</td>
</tr>
```

```
</table>

</div>

</section>

</div>



<footer>
    <p>© 2026 SkyFly Online Flight Reservation System</p>
</footer>



<script>

let selectedFlight = null;
let booking = null;
let bookings = 0;
let revenue = 0;

const flights = [
{
    number:"SF101",
    from:"Chennai",
    to:"Delhi",
    time:"06:30 AM",
    seats:25,
    fare:5500
},
{
    number:"SF202",
    from:"Chennai",
    to:"Mumbai",
    time:"09:00 AM",
    seats:18,
    fare:4800
},
{
    number:"SF303",
    from:"Bangalore",
    to:"Delhi",
    time:"01:30 PM",
    seats:30,
    fare:6200
},
{
    number:"SF404",
    from:"Delhi",
    to:"Mumbai",
    time:"05:00 PM",
    seats:20,
    fare:4500
}
];



function showPage(page){

    document.querySelectorAll(".page").forEach(function(p){
        p.classList.remove("active");
    });

    document.getElementById(page).classList.add("active");

    window.scrollTo(0,0);
}



function register(){

    let name=document.getElementById("regName").value;
    let email=document.getElementById("regEmail").value;
    let password=document.getElementById("regPassword").value;

    if(name==="" || email==="" || password===""){
        document.getElementById("registerMsg").innerHTML="Please fill all fields.";
        return;
    }

    localStorage.setItem("userName",name);
```

```
    localStorage.setItem("userEmail",email);
    localStorage.setItem("userPassword",password);

    document.getElementById("registerMsg").innerHTML=
    "Registration successful!";

}

function login(){

    let email=document.getElementById("loginEmail").value;
    let password=document.getElementById("loginPassword").value;

    if(
        email===localStorage.getItem("userEmail") &&
        password===localStorage.getItem("userPassword")
    ){

        document.getElementById("loginMsg").innerHTML="Login successful!";
        showPage("search");

    }else{

        document.getElementById("loginMsg").innerHTML=
        "Invalid email or password.";

    }

}

function resetPassword(){

    let email=document.getElementById("forgotEmail").value;

    if(email===localStorage.getItem("userEmail")){

        document.getElementById("forgotMsg").innerHTML=
        "Password reset link sent to your email.";

    }else{

        document.getElementById("forgotMsg").innerHTML=
        "Email not found.";

    }

}

function searchFlights(){

    let from=document.getElementById("from").value;
    let to=document.getElementById("to").value;

    let results=flights.filter(function(f){
        return f.from===from && f.to===to;
    });

    let output="";

    if(results.length===0){

        output="<div class='card'><h3>No flights available.</h3></div>";

    }else{

        results.forEach(function(f){

            output+=`

            <div class="flight">
```

```
            <h3>✈ ${f.number}</h3>

            <p>${f.from} → ${f.to}</p>

            <p>Departure: ${f.time}</p>

            <p>Available Seats: ${f.seats}</p>

            <p>Fare: ₹${f.fare}</p>

            <button onclick="bookFlight('${f.number}')">
            Book Ticket
            </button>

            <br><br>

            </div>

            `;

        });

    }

    document.getElementById("flightResults").innerHTML=output;

}



function bookFlight(number){

    selectedFlight=flights.find(function(f){
        return f.number===number;
    });

    document.getElementById("selectedFlight").innerHTML=`

        <div class="message">
        <b>${selectedFlight.number}</b><br>
        ${selectedFlight.from} → ${selectedFlight.to}<br>
        Departure: ${selectedFlight.time}<br>
        Fare: ₹${selectedFlight.fare}
        </div>

    `;

    showPage("booking");

}



function proceedPayment(){

    let name=document.getElementById("passengerName").value;
    let age=document.getElementById("passengerAge").value;

    if(name==="" || age===""){

        alert("Please enter passenger details.");
        return;

    }

    document.getElementById("paymentAmount").innerHTML=
    "Total Amount: ₹"+selectedFlight.fare;

    showPage("payment");

}



function makePayment(){

    booking={
```

```
        id:"SF"+Math.floor(Math.random()*90000+10000),

        passenger:document.getElementById("passengerName").value,

        flight:selectedFlight.number,

        route:selectedFlight.from+" → "+selectedFlight.to,

        seat:document.getElementById("seat").value,

        amount:selectedFlight.fare,

        status:"Confirmed"

    };

    bookings++;

    revenue+=selectedFlight.fare;

    selectedFlight.seats--;

    document.getElementById("paymentMsg").innerHTML=
    "Payment successful! Ticket booked.";

    document.getElementById("totalBookings").innerHTML=bookings;

    document.getElementById("revenue").innerHTML="₹"+revenue;

    document.getElementById("ticketDetails").innerHTML=`

        <h3>✈ Booking Confirmed</h3>

        <p><b>Booking ID:</b> ${booking.id}</p>
        <p><b>Passenger:</b> ${booking.passenger}</p>
        <p><b>Flight:</b> ${booking.flight}</p>
        <p><b>Route:</b> ${booking.route}</p>
        <p><b>Seat:</b> ${booking.seat}</p>
        <p><b>Amount:</b> ₹${booking.amount}</p>
        <p><b>Status:</b> ${booking.status}</p>

    `;

    showPage("myticket");

}



function cancelTicket(){

    if(!booking){

        alert("No ticket available.");
        return;

    }

    booking.status="Cancelled";

    document.getElementById("ticketDetails").innerHTML+=
    "<p style='color:red'><b>Ticket Cancelled</b></p>";

}



function printTicket(){

    if(!booking){

        alert("No ticket available.");
        return;

    }
```

```
    window.print();

}



function checkStatus(){

    let id=document.getElementById("bookingId").value;

    if(booking && id===booking.id){

        document.getElementById("statusResult").innerHTML=
        "Booking Status: "+booking.status;

    }else{

        document.getElementById("statusResult").innerHTML=
        "Booking not found.";

    }

}



function submitFeedback(){

    let name=document.getElementById("feedbackName").value;
    let feedback=document.getElementById("feedbackText").value;

    if(name==="" || feedback===""){

        alert("Please enter feedback.");
        return;

    }

    document.getElementById("feedbackMsg").innerHTML=
    "Thank you for your valuable feedback!";

}



function adminLogin(){

    let user=document.getElementById("adminUser").value;
    let pass=document.getElementById("adminPass").value;

    if(user==="admin" && pass==="admin123"){

        document.getElementById("adminMsg").innerHTML=
        "Admin login successful!";

        setTimeout(function(){
            showPage("adminPanel");
        },500);

    }else{

        document.getElementById("adminMsg").innerHTML=
        "Invalid admin credentials.";

    }

}



function addFlight(){

    let number=document.getElementById("newFlight").value;
    let route=document.getElementById("newRoute").value;
    let seats=document.getElementById("newSeats").value;
    let fare=document.getElementById("newFare").value;
```

```
    if(number==="" || route==="" || seats==="" || fare===""){

        alert("Please enter all flight details.");
        return;

    }

    document.getElementById("flightTable").innerHTML+=`

        <tr>
        <td>${number}</td>
        <td>${route}</td>
        <td>${seats}</td>
        <td>₹${fare}</td>
        </tr>

    `;

    document.getElementById("totalFlights").innerHTML=
    parseInt(document.getElementById("totalFlights").innerHTML)+1;

}

</script>

</body>
</html>
```

4. INPUT

The program uses the following inputs:

- Registration: Full name, email address, and password.
- Login and password reset: Email address and password, or registered email address for the reset message.
- Flight search: Origin city, destination city, journey date, and travel class (Economy, Business, or First Class).
- Passenger booking: Passenger name, age, passport/ID number, and selected seat.
- Payment: Payment method (Credit Card, Debit Card, UPI, or Net Banking), card/UPI number, and PIN/CVV fields.
- Booking status: Booking ID.
- Feedback: Customer name, rating, and feedback text.
- Administrator: Username and password; the code checks the sample credentials admin and admin123.
- Manage flights: Flight number, route, seat count, and fare.

Sample flight data defined in the JavaScript includes SF101 (Chennai–Delhi, ₹5500), SF202 (Chennai–Mumbai, ₹4800), SF303 (Bangalore–Delhi, ₹6200), and SF404 (Delhi–Mumbai, ₹4500).

5. OUTPUT

The webpage displays a SkyFly Reservation interface with navigation, a home page and feature list, flight-search controls, and flight results. Other pages provide customer login and registration, booking, payment, ticket details, feedback, administrator tools, and reports.

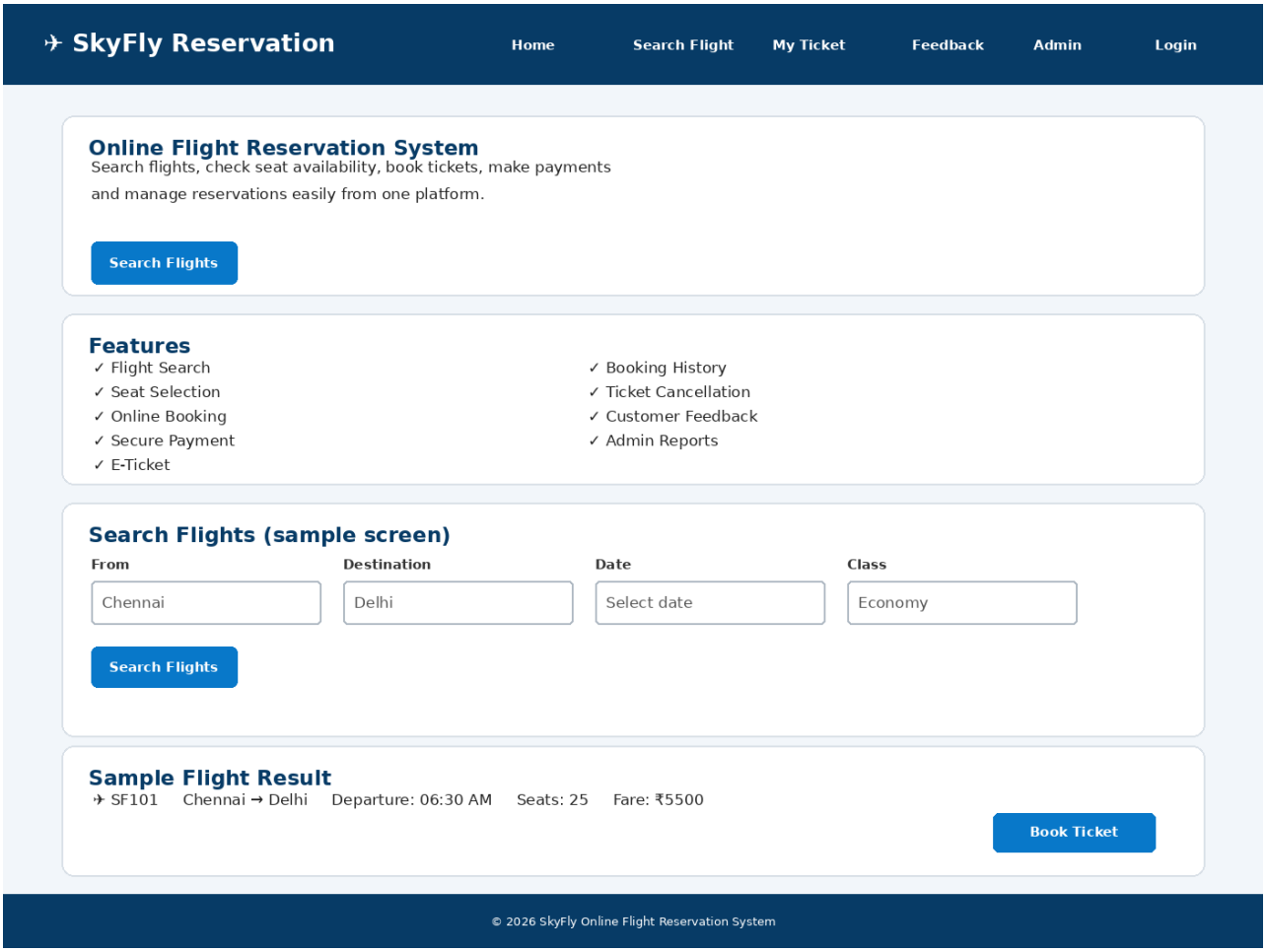


Figure 1: Illustrative output screen based on the supplied HTML/CSS design.

6. RESULT

Thus, the Online Flight Reservation System front-end was prepared using HTML, CSS, and JavaScript. The program demonstrates flight searching, simulated booking and payment, ticket display, cancellation, feedback, and basic administrator functions using sample data and browser-side storage.

Limitation: This is a client-side demonstration. Registration data is stored in localStorage, the password-reset message is simulated, and payment is not connected to a real payment gateway. A production system would require secure server-side authentication, a database, and a verified payment service.

ONLINE CAKE ORDERING SYSTEM

1. AIM

To design and develop an Online Cake Ordering System using HTML, CSS, and JavaScript. The system provides cake browsing and filtering, product descriptions, a shopping cart, shipping-address entry, simulated payment and order placement, order tracking, customer registration/login, and basic administrator functions.

2. PROCEDURE

1. Open a text editor or IDE such as Visual Studio Code.
2. Create a new file and save the complete source code as index.html.
3. Build the page structure using HTML sections for Home, Registration, Login, Cakes, Cake Details, Cart, Shipping, Payment, Orders, Tracking, and Admin pages.
4. Apply CSS to style the header, navigation, cards, product grid, forms, buttons, tables, and responsive layout.
5. Define the sample cake objects in JavaScript, including each cake's name, category, subcategory, price, image, and description.
6. Implement page navigation and cake search/filter functions based on name, category, subcategory, and price.
7. Implement viewing cake details, adding items to the cart, removing cart items, and calculating the cart total.
8. Collect shipping details and implement a simulated payment/order placement process that generates an order ID.
9. Display order history and status, and provide administrator login, product management, and order-status management.
10. Save the file and open it in a browser. Test the sample product cards, filters, cart, shipping, order placement, tracking, and admin functions.

3. PROGRAM CODE

Complete HTML, CSS, and JavaScript source code from the supplied file:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Online Cake Ordering System</title>

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial,sans-serif;
}

body{
    background:#fff5f7;
    color:#333;
}

header{
    background:#d6336c;
    color:white;
    padding:18px 40px;
    display:flex;
```

```
    justify-content:space-between;
    align-items:center;
}

header h1{
    font-size:25px;
}

nav button{
    background:none;
    border:none;
    color:white;
    margin:5px;
    cursor:pointer;
    font-size:14px;
}

.container{
    width:90%;
    max-width:1200px;
    margin:30px auto;
}

.page{
    display:none;
}

.active{
    display:block;
}

.card{
    background:white;
    padding:25px;
    margin-bottom:20px;
    border-radius:12px;
    box-shadow:0 3px 12px #ddd;
}

h2{
    color:#d6336c;
    margin-bottom:20px;
}

input,select,textarea{
    width:100%;
    padding:12px;
    margin:8px 0 15px;
    border:1px solid #ccc;
    border-radius:6px;
}

button{
    background:#d6336c;
    color:white;
    border:none;
    padding:11px 18px;
    border-radius:6px;
    cursor:pointer;
}

button:hover{
    background:#ad2455;
}

.hero{
    text-align:center;
    padding:60px 20px;
    background:#ffe0e9;
    border-radius:15px;
}

.hero h2{
    font-size:35px;
}

.filters{
    display:grid;
    grid-template-columns:repeat(4,1fr);
    gap:15px;
}

.cakes{
    display:grid;
    grid-template-columns:repeat(3,1fr);
    gap:20px;
```

```
}

.cake{
    background:white;
    border-radius:12px;
    overflow:hidden;
    box-shadow:0 3px 12px #ddd;
}

.cake img{
    width:100%;
    height:200px;
    object-fit:cover;
}

.cake-content{
    padding:18px;
}

.cake-content h3{
    color:#d6336c;
    margin-bottom:8px;
}

.cake-content button{
    margin-top:10px;
}

.cart-item{
    border-bottom:1px solid #ddd;
    padding:15px 0;
}

.price{
    color:#d6336c;
    font-weight:bold;
}

table{
    width:100%;
    border-collapse:collapse;
}

th,td{
    padding:12px;
    border:1px solid #ddd;
    text-align:center;
}

th{
    background:#d6336c;
    color:white;
}

.status{
    padding:7px 12px;
    background:#ffe0e9;
    color:#d6336c;
    border-radius:15px;
}

footer{
    background:#d6336c;
    color:white;
    text-align:center;
    padding:20px;
    margin-top:40px;
}

@media(max-width:800px){

    .cakes{
        grid-template-columns:repeat(2,1fr);
    }

    .filters{
        grid-template-columns:repeat(2,1fr);
    }

    header{
        flex-direction:column;
        gap:15px;
    }
```

```
}

@media(max-width:550px){

    .cakes{
        grid-template-columns:1fr;
    }

    .filters{
        grid-template-columns:1fr;
    }

}

</style>
</head>

<body>

<header>

<h1>🎂 SweetCake</h1>

<nav>
<button onclick="showPage('home')">Home</button>
<button onclick="showPage('cakes')">Cakes</button>
<button onclick="showPage('cart')">Cart 🛒</button>
<button onclick="showPage('orders')">My Orders</button>
<button onclick="showPage('login')">Login</button>
<button onclick="showPage('adminLogin')">Admin</button>
</nav>

</header>



<div class="container">



<!-- HOME -->

<section id="home" class="page active">

<div class="hero">

<h2>Delicious Cakes for Every Occasion 🎂</h2>

<p>
Freshly prepared cakes delivered to your doorstep.
</p>

<br>

<button onclick="showPage('cakes')">
Shop Cakes
</button>

</div>

<br>

<div class="card">

<h2>Why Choose SweetCake?</h2>

<p>🍰 Wide variety of cakes</p>
<p>🎉 Cakes for every occasion</p>
<p>🚚 Fast delivery</p>
<p>💳 Easy online payment</p>
<p>📦 Order tracking</p>

</div>

</section>
```

```
<!-- REGISTER -->

<section id="register" class="page">

<div class="card">

<h2>Student Registration</h2>

<input id="regName" placeholder="Full Name">

<input id="regEmail" type="email" placeholder="Email">

<input id="regPassword" type="password" placeholder="Password">

<button onclick="register()">
Register
</button>

<p id="registerMsg"></p>

</div>

</section>



<!-- LOGIN -->

<section id="login" class="page">

<div class="card">

<h2>Student Login</h2>

<input id="loginEmail" type="email" placeholder="Email">

<input id="loginPassword" type="password" placeholder="Password">

<button onclick="login()">
Login
</button>

<button onclick="showPage('register')">
Register
</button>

<p id="loginMsg"></p>

</div>

</section>



<!-- CAKES -->

<section id="cakes" class="page">

<div class="card">

<h2>Search & Filter Cakes</h2>

<div class="filters">

<input
id="search"
placeholder="Search Cake"
onkeyup="displayCakes()">

<select id="category" onchange="displayCakes()">

<option value="All">All Categories</option>
<option value="Birthday">Birthday</option>
<option value="Wedding">Wedding</option>
<option value="Anniversary">Anniversary</option>
<option value="Chocolate">Chocolate</option>
```

```
</select>



<select id="subcategory" onchange="displayCakes()">


<option value="All">All Sub Categories</option>
<option value="Eggless">Eggless</option>
<option value="Cream">Cream</option>
<option value="Fruit">Fruit</option>


</select>



<select id="price" onchange="displayCakes()">


<option value="All">All Prices</option>
<option value="500">Below ₹500</option>
<option value="1000">Below ₹1000</option>
<option value="2000">Below ₹2000</option>


</select>


</div>


</div>


<div id="cakeList" class="cakes"></div>


</section>



<!-- CAKE DETAILS -->


<section id="details" class="page">


<div class="card">


<h2>Cake Description</h2>


<div id="cakeDetails"></div>


<button onclick="showPage('cakes')">
Back to Cakes
</button>


</div>


</section>



<!-- CART -->


<section id="cart" class="page">


<div class="card">


<h2>Shopping Cart 🛒</h2>


<div id="cartItems"></div>


<h3 id="cartTotal"></h3>


<br>


<button onclick="showPage('shipping')">
Place Order
</button>


</div>


</section>
```

```
<!-- SHIPPING -->

<section id="shipping" class="page">

<div class="card">

<h2>Shipping Address</h2>

<input id="addressName" placeholder="Name">

<input id="addressPhone" placeholder="Phone Number">

<textarea
id="address"
placeholder="Enter Shipping Address"></textarea>

<input id="city" placeholder="City">

<input id="pincode" placeholder="Pincode">

<button onclick="saveAddress()">
Save / Update Address
</button>

<p id="addressMsg"></p>

<br>

<button onclick="showPage('payment')">
Continue to Payment
</button>

</div>

</section>



<!-- PAYMENT -->

<section id="payment" class="page">

<div class="card">

<h2>Payment</h2>

<h3 id="paymentTotal"></h3>

<select id="paymentMethod">

<option>UPI</option>
<option>Credit Card</option>
<option>Debit Card</option>
<option>Net Banking</option>

</select>

<input placeholder="Payment Number">

<input type="password" placeholder="PIN / CVV">

<button onclick="placeOrder()">
Pay & Place Order
</button>

<p id="paymentMsg"></p>

</div>

</section>



<!-- ORDERS -->

<section id="orders" class="page">
```

```
<div class="card">

<h2>My Orders</h2>

<div id="orderList">

<p>No orders yet.</p>

</div>

</div>

</section>


<!-- SHIPPING STATUS -->

<section id="shippingStatus" class="page">

<div class="card">

<h2>View Shipping Status</h2>

<input id="trackId" placeholder="Enter Order ID">

<button onclick="trackOrder()">
Track Order
</button>

<p id="trackResult"></p>

</div>

</section>


<!-- ADMIN LOGIN -->

<section id="adminLogin" class="page">

<div class="card">

<h2>Admin Login</h2>

<input id="adminUser" placeholder="Username">

<input id="adminPassword"
type="password"
placeholder="Password">

<button onclick="adminLogin()">
Login
</button>

<p id="adminMsg"></p>

</div>

</section>


<!-- ADMIN PANEL -->

<section id="adminPanel" class="page">

<div class="card">

<h2>Admin Panel</h2>

<button onclick="showPage('manageProducts')">
Manage Products
</button>
```

```
<button onclick="showPage('manageOrders')">
Manage Orders
</button>

<button onclick="adminLogout()">
Logout
</button>

</div>

</section>



<!-- MANAGE PRODUCTS -->

<section id="manageProducts" class="page">

<div class="card">

<h2>Manage Products</h2>

<input id="productName"
placeholder="Cake Name">

<input id="productPrice"
type="number"
placeholder="Price">

<select id="productCategory">

<option>Birthday</option>
<option>Wedding</option>
<option>Anniversary</option>
<option>Chocolate</option>

</select>

<input id="productImage"
placeholder="Image URL">

<textarea
id="productDescription"
placeholder="Cake Description"></textarea>

<button onclick="addProduct()">
Post Cake
</button>

<br><br>

<table>

<thead>

<tr>
<th>Product</th>
<th>Category</th>
<th>Price</th>
</tr>

</thead>

<tbody id="productTable"></tbody>

</table>

</div>

</section>



<!-- MANAGE ORDERS -->

<section id="manageOrders" class="page">
```

```
<div class="card">

<h2>Manage Orders</h2>

<table>

<thead>

<tr>
<th>Order ID</th>
<th>Customer</th>
<th>Total</th>
<th>Status</th>
<th>Update</th>
</tr>

</thead>

<tbody id="adminOrders"></tbody>

</table>

</div>

</section>



</div>



<footer>

<p>© 2026 SweetCake - Online Cake Ordering System</p>

</footer>



<script>



// CAKE DATA

let cakes = [

{
id:1,
name:"Chocolate Truffle",
category:"Chocolate",
sub:"Cream",
price:650,
image:"https://images.unsplash.com/photo-1578985545062-69928b1d9587",
description:"Rich and delicious chocolate truffle cake."
},

{
id:2,
name:"Birthday Special",
category:"Birthday",
sub:"Cream",
price:550,
image:"https://images.unsplash.com/photo-1551024506-0bccd828d307",
description:"Beautiful birthday cake with fresh cream."
},

{
id:3,
name:"Strawberry Cake",
category:"Anniversary",
sub:"Fruit",
price:750,
image:"https://images.unsplash.com/photo-1565958011703-44f9829ba187",
description:"Fresh strawberry cake perfect for anniversaries."
},

{
id:4,
name:"Wedding White Cake",
category:"Wedding",
sub:"Cream",
price:1500,
```

```
image:"https://images.unsplash.com/photo-1535254973040-607b474cb50d",
description:"Elegant white cake for weddings."
},


{
id:5,
name:"Eggless Vanilla",
category:"Birthday",
sub:"Eggless",
price:450,
image:"https://images.unsplash.com/photo-1464349095431-e9a21285b5f3",
description:"Soft and delicious eggless vanilla cake."
},


{
id:6,
name:"Fruit Celebration",
category:"Anniversary",
sub:"Fruit",
price:900,
image:"https://images.unsplash.com/photo-1602351447937-745cb720612f",
description:"Fresh fruit cake with delicious toppings."
}


];



let cart=[];


let orders=[];


let address="";



// PAGE NAVIGATION


function showPage(page){


document.querySelectorAll(".page").forEach(function(p){


p.classList.remove("active");


});


document.getElementById(page).classList.add("active");


if(page==="cakes"){
displayCakes();
}


if(page==="cart"){
displayCart();
}


if(page==="orders"){
displayOrders();
}


if(page==="payment"){
calculatePayment();
}


if(page==="manageOrders"){
displayAdminOrders();
}


}



// REGISTER


function register(){


let name=document.getElementById("regName").value;


let email=document.getElementById("regEmail").value;


let password=document.getElementById("regPassword").value;
```

```
if(!name || !email || !password){

alert("Please fill all fields");

return;

}

localStorage.setItem("cakeUserName",name);

localStorage.setItem("cakeUserEmail",email);

localStorage.setItem("cakeUserPassword",password);

document.getElementById("registerMsg").innerHTML=
"Registration successful!";

}



// LOGIN

function login(){

let email=document.getElementById("loginEmail").value;

let password=document.getElementById("loginPassword").value;

if(
email===localStorage.getItem("cakeUserEmail") &&
password===localStorage.getItem("cakeUserPassword")
){

document.getElementById("loginMsg").innerHTML=
"Login successful!";

showPage("cakes");

}

else{

document.getElementById("loginMsg").innerHTML=
"Invalid email or password.";

}

}



// DISPLAY CAKES

function displayCakes(){

let search=document.getElementById("search").value.toLowerCase();

let category=document.getElementById("category").value;

let sub=document.getElementById("subcategory").value;

let price=document.getElementById("price").value;

let result=cakes.filter(function(cake){

return (

cake.name.toLowerCase().includes(search)

&&

(category==="All" || cake.category===category)
```

```
&&

(sub==="All" || cake.sub===sub)

&&

(price==="All" || cake.price<=Number(price))

);

});


let output="";


result.forEach(function(cake){

output+=`

<div class="cake">

<img src="${cake.image}">

<div class="cake-content">

<h3>${cake.name}</h3>

<p>${cake.description}</p>

<p class="price">₹${cake.price}</p>

<button onclick="viewCake(${cake.id})">
View Description
</button>

<button onclick="addToCart(${cake.id})">
Add to Cart
</button>

</div>

</div>

`;

});


document.getElementById("cakeList").innerHTML=
output || "<p>No cakes found.</p>";

}


// VIEW CAKE

function viewCake(id){

let cake=cakes.find(c=>c.id===id);

document.getElementById("cakeDetails").innerHTML=`

<img src="${cake.image}"
style="width:300px;max-width:100%;border-radius:10px;">

<h3>${cake.name}</h3>

<p>${cake.description}</p>

<h3>Price: ₹${cake.price}</h3>

<br>
```

```
<button onclick="addToCart(${cake.id})">
Add to Cart
</button>

`;

showPage("details");

}


// ADD CART

function addToCart(id){

let cake=cakes.find(c=>c.id===id);

cart.push(cake);

alert(cake.name+" added to cart!");

}


// DISPLAY CART

function displayCart(){

let output="";

let total=0;


cart.forEach(function(cake,index){

total+=cake.price;

output+=`

<div class="cart-item">

<h3>${cake.name}</h3>

<p>₹${cake.price}</p>

<button onclick="removeCart(${index})">
Remove
</button>

</div>

`;

});


document.getElementById("cartItems").innerHTML=
output || "<p>Your cart is empty.</p>";

document.getElementById("cartTotal").innerHTML=
"Total: ₹"+total;

}


// REMOVE CART

function removeCart(index){

cart.splice(index,1);
```

```
displayCart();

}


// SHIPPING ADDRESS

function saveAddress(){

let name=document.getElementById("addressName").value;

let phone=document.getElementById("addressPhone").value;

let addr=document.getElementById("address").value;

let city=document.getElementById("city").value;

let pincode=document.getElementById("pincode").value;


if(!name || !phone || !addr || !city || !pincode){

alert("Please enter complete shipping address.");

return;

}


address=
name+", "+phone+", "+addr+", "+city+" - "+pincode;


document.getElementById("addressMsg").innerHTML=
"Shipping address saved successfully.";

}


// PAYMENT TOTAL

function calculatePayment(){

let total=cart.reduce(function(sum,cake){

return sum+cake.price;

},0);


document.getElementById("paymentTotal").innerHTML=
"Total Amount: ₹"+total;

}


// PLACE ORDER

function placeOrder(){

if(cart.length===0){

alert("Cart is empty.");

return;

}

if(address===""){

alert("Please add shipping address.");
```

```
return;

}


let total=cart.reduce(function(sum,cake){

return sum+cake.price;

},0);


let order={

id:"ORD"+Math.floor(Math.random()*90000+10000),

items:[...cart],

total:total,

address:address,

status:"Order Placed"

};


orders.push(order);

cart=[];


document.getElementById("paymentMsg").innerHTML=
"Payment successful! Order placed.";

displayOrders();

setTimeout(function(){

showPage("orders");

},800);

}


// DISPLAY ORDERS

function displayOrders(){

let output="";


orders.forEach(function(order){

output+=`

<div class="card">

<h3>Order ID: ${order.id}</h3>

<p>Total: ₹${order.total}</p>

<p>Shipping Address: ${order.address}</p>

<p>

Status:

<span class="status">
```

```
${order.status}
</span>

</p>

<br>

<button onclick="trackSpecific('${order.id}')">
View Shipping Status
</button>

</div>

`;

});



document.getElementById("orderList").innerHTML=
output || "<p>No orders yet.</p>";

}



// TRACK ORDER

function trackSpecific(id){

document.getElementById("trackId").value=id;

showPage("shippingStatus");

trackOrder();

}



function trackOrder(){

let id=document.getElementById("trackId").value;

let order=orders.find(o=>o.id===id);



if(order){

document.getElementById("trackResult").innerHTML=

"Order <b>"+order.id+
"</b> Status: <b>"+order.status+"</b>";

}

else{

document.getElementById("trackResult").innerHTML=
"Order not found.";

}

}



// ADMIN LOGIN

function adminLogin(){

let user=document.getElementById("adminUser").value;

let pass=document.getElementById("adminPassword").value;
```

```
if(user==="admin" && pass==="admin123"){

document.getElementById("adminMsg").innerHTML=
"Admin login successful!";

setTimeout(function(){

showPage("adminPanel");

},500);

}

else{

document.getElementById("adminMsg").innerHTML=
"Invalid admin username or password.";

}

}


// ADMIN LOGOUT

function adminLogout(){

showPage("home");

}


// ADD PRODUCT

function addProduct(){

let name=document.getElementById("productName").value;

let price=document.getElementById("productPrice").value;

let category=document.getElementById("productCategory").value;

let image=document.getElementById("productImage").value;

let description=
document.getElementById("productDescription").value;


if(!name || !price || !image || !description){

alert("Please enter all product details.");

return;

}


let product={

id:cakes.length+1,

name:name,

category:category,

sub:"Cream",

price:Number(price),

image:image,

description:description
```

```
};

cakes.push(product);

document.getElementById("productTable").innerHTML+=`
<tr>
<td>${name}</td>
<td>${category}</td>
<td>₹${price}</td>
</tr>
`;

alert("Cake added successfully!");
}

// ADMIN ORDERS
function displayAdminOrders(){
let output="";

orders.forEach(function(order,index){
output+=`
<tr>
<td>${order.id}</td>
<td>Customer</td>
<td>₹${order.total}</td>
<td>${order.status}</td>
<td>
<select onchange="updateOrder(${index},this.value)">
<option>Order Placed</option>
<option>Preparing</option>
<option>Out for Delivery</option>
<option>Delivered</option>
<option>Cancelled</option>
</select>
</td>
</tr>
`;
});
```

```
document.getElementById("adminOrders").innerHTML=
output || `

<tr>

<td colspan="5">
No orders available
</td>

</tr>

`;

}



// UPDATE ORDER STATUS

function updateOrder(index,status){

orders[index].status=status;

alert("Order status updated to "+status);

}



// INITIAL DISPLAY

displayCakes();

</script>

</body>
</html>
```

4. INPUT

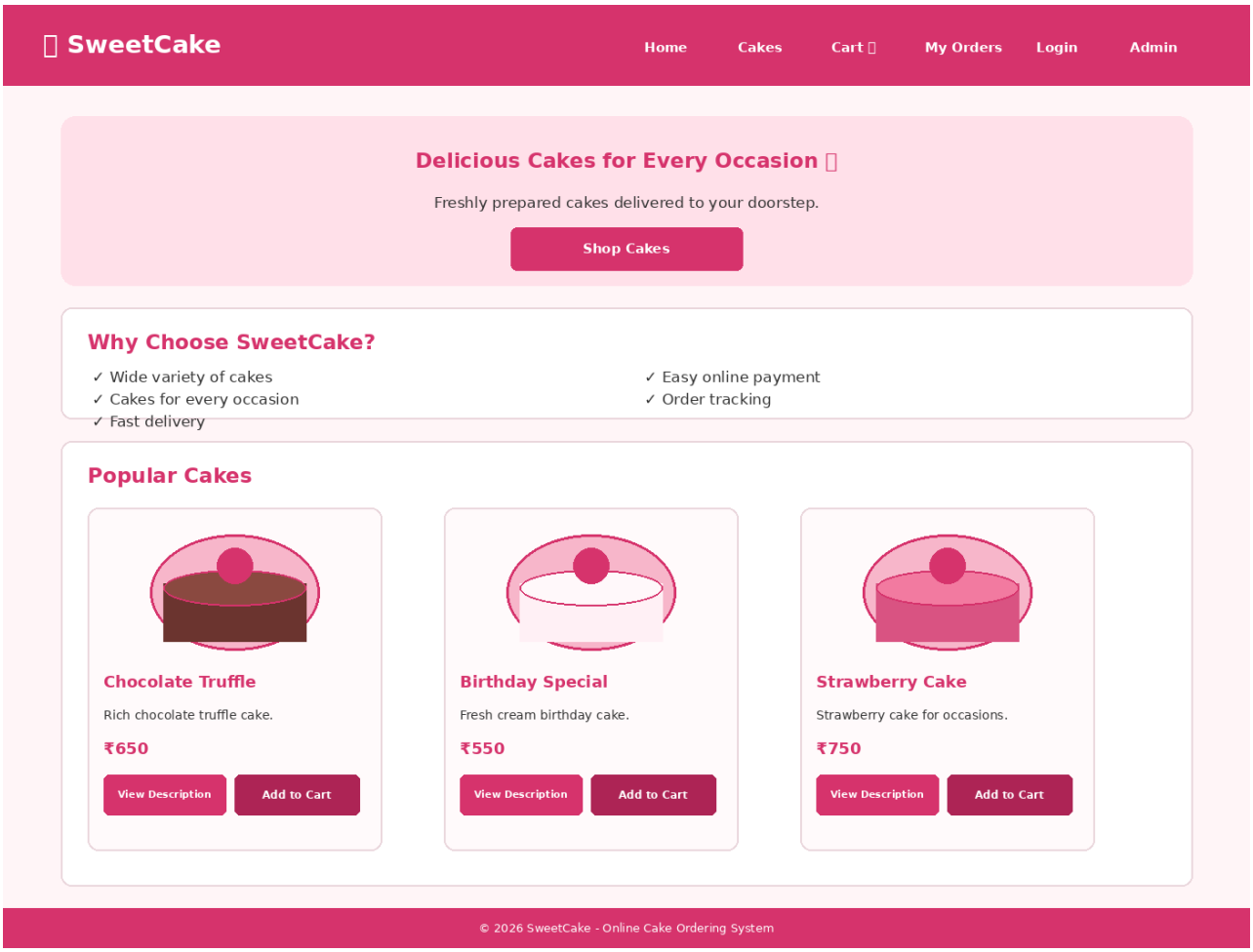
The program accepts the following inputs:

- Customer registration: Full name, email, and password.
- Customer login: Email and password.
- Cake search and filters: Search text, category, subcategory, and price range.
- Shipping address: Name, phone number, street/address, city, and pincode.
- Payment: Payment method (UPI, Credit Card, Debit Card, or Net Banking), payment number, and PIN/CVV fields.
- Administrator login: Username and password. The sample credentials in the code are username admin and password admin123.
- Product management: Cake name, price, category, image URL, and description.
- Order management: A selected order status such as Order Placed, Preparing, Out for Delivery, Delivered, or Cancelled.

Sample cakes include Chocolate Truffle (₹650), Birthday Special (₹550), Strawberry Cake (₹750), Wedding White Cake (₹1500), Eggless Vanilla (₹450), and Fruit Celebration (₹900).

5. OUTPUT

The interface displays the SweetCake home page, cake catalog with search and filters, cake descriptions, shopping cart, shipping form, payment page, order history and tracking, and administrator pages for managing products and orders.



6. RESULT

Thus, the Online Cake Ordering System interface was developed using HTML, CSS, and JavaScript. The program demonstrates cake filtering, product viewing, cart operations, shipping-address entry, simulated order placement, order tracking, and basic administrator product/order management.

Limitation: This is a browser-side demonstration. User credentials and order information are not managed by a secure backend, and the payment step is simulated rather than connected to a real payment gateway. A production system requires secure server-side authentication, database storage, validated order processing, and a payment provider.